Applied Linear Algebra

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Outline

Matrices

Special matrices

Addition and scalar multiplication of matrices

Matrix-vector multiplication

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Matrices

- a matrix is a rectangular array of numbers, e.g.

$$\begin{bmatrix} 1.8 & -1 & 2 & 1.3 \\ 0 & 1 & 3.2 & -5.4 \\ 1.2 & -1 & 2 & 2.2 \end{bmatrix} \qquad \textit{or} \qquad \begin{pmatrix} 1.8 & -1 & 2 & 1.3 \\ 0 & 1 & 3.2 & -5.4 \\ 1.2 & -1 & 2 & 2.2 \end{pmatrix}$$

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- every matrix is characterized by its size i.e: (number of rows) $\times$ (number of columns) usually denoted by $m \times n$; matrix above is 3×4.

- elements also called entries or coefficients

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General form of a matrix

- A $m \times n$ matrix A is given by:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

- a_{ij} is i, j element of matrix A: i is the row index, j is the column index, e.g: below $a_{23} =?$

$$\begin{bmatrix} 1.8 & -1 & 2 & 1.3 \\ 0 & 1 & 3.2 & -5.4 \\ 1.2 & -1 & 2 & 2.2 \end{bmatrix}$$

- What are a_{12}, a_{34}?

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- What are a_{12}, a_{34}?
- indexes start at 1; in computer languages indexes often start at 0.

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Matrix shapes

An $m \times n$ matrix A

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Matrix shapes

An $m \times n$ matrix A

- tall if $m > n$ e.g
- wide if $m < n$ e.g
- square if $m = n$ e.g

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Column and row vectors

- we consider an $n \times 1$ matrix to be an n-vector
- we consider a 1×1 matrix to be a number
- a $1 \times n$ matrix is called a *row vector*, e.g.,

$$\begin{bmatrix} 1.2 & 0.3 & 2.5 & 1.7 \end{bmatrix}$$

which is not the same (in matrix notations) as

$$\begin{bmatrix} 1.2 \\ 0.3 \\ 2.5 \\ 1.7 \end{bmatrix}$$

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Columns and rows of a matrix A

Suppose A is an $m \times n$ matrix with entries a_{ij} for $i = 1, \ldots, m, j = 1, \ldots, n$

- its jth column is (the m-vector)

$$A_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \ldots \\ a_{mj} \end{bmatrix}$$

- its ith row is (the n-vector)

$$R_j = \begin{bmatrix} a_{i1} & a_{i2} & \ldots & a_{in} \end{bmatrix}$$

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- As a specific example the 2×3 matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ has first row $R_1 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and second column $A_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

Column and row representation of matrix

A is an $m \times n$ matrix

- can express as block matrix with its (n-vector) columns

$$A = \left[A_1 \mid \ldots \mid A_n\right]$$

eg

$$A = \left[\begin{array}{c|c|c} 2 & 3 & 1 \\ 1 & 0 & 3 \end{array}\right] = \left[A_1 \mid A_2 \mid A_3\right]$$

- can express as block matrix with its (m-vector) rows

$$A = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix} \qquad A = \left[\begin{array}{ccc} 2 & 3 & 1 \\ \hline 1 & 0 & 3 \end{array}\right] = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

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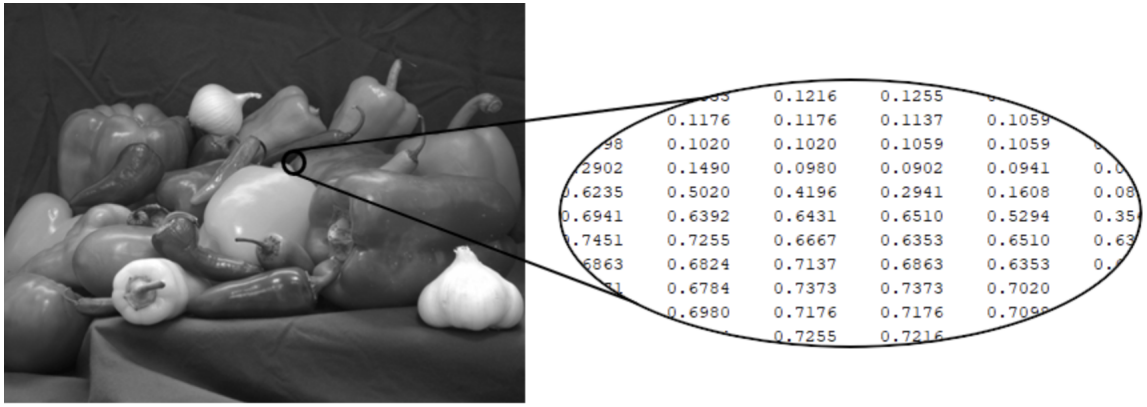
Examples of matrices

- image: x_{ij} is i, j pixel value in a greyscale image



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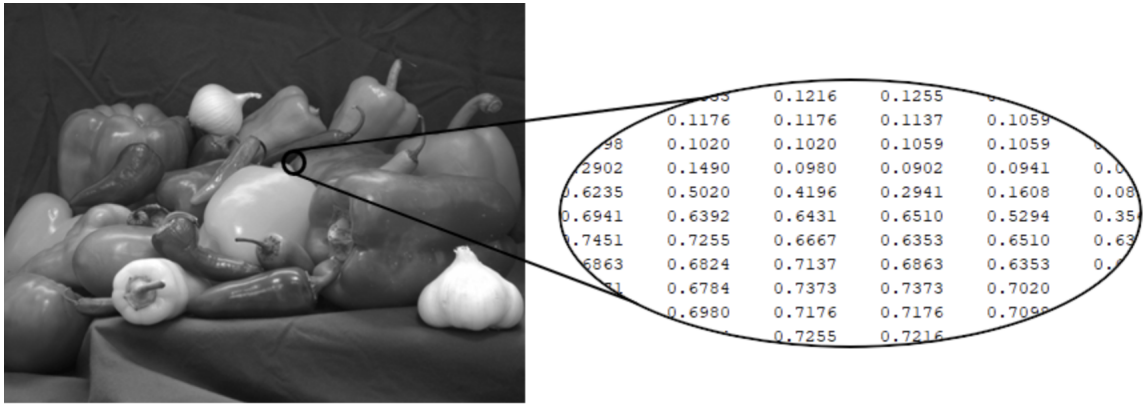

- prices(in SAR) from multiple supplier: p_{ij} is the price that vendor j charges for good i

	Vendor 1	Vendor 2	Vendor 3	Vendor 4	Vendor 5
apple	10.95	9.5	8.95	9.45	9.95
banana	17.75	19.5	18.95	18.5	18.25
eggs	25.4	26.95	25.85	26.25	25.95
tomatoes	4.75	5.5	4.25	5.95	4.95
lettuce	2.95	3.30	3.10	3.45	3.25
plain yogurt (2kg)	14.75	15.5	14.95	16.25	15.10

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- Identity matrix is square matrix with $I_{ii} = 1$ and $I_{ij} = 0$ for i, j, e.g.,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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- sparse matrix: most entries are zero:
  - examples: 0 and I
  - can be stored and manipulated efficiently
  - **nnz**(A) is number of nonzero entries

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Diagonal and triangular matrices:

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- diagonal matrix:
- lower triangular matrix:
- upper triangular matrix:

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Transpose

- the transpose of an $m \times n$ matrix A is denoted A^T and defined by converting column to row vectors (and vice versa)
- for example, $A = \begin{bmatrix} 0 & 3 \\ 6 & 1 \\ 2 & -1 \end{bmatrix}$ then $A^T =$

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- the transpose of a single row vector is a column vector.
- symmetric matrix: $A = A^T$ eg.

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Matrix addition

- (just like vectors) we can add or subtract matrices of the same size:

$$
\begin{aligned}
A + B &= \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & b_{mn} \end{bmatrix} \\
&= \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \dots & a_{mn} + b_{mn} \end{bmatrix}.
\end{aligned}
$$

- (subtraction is similar)

Scalar multiplication of a matrix

- Let A be a $m \times n$ matrix and $k \in \mathbb{R}$. Then, kA is defined as

$$kA = \begin{bmatrix} ka_{11} & \dots & ka_{1n} \\ \vdots & \ddots & \vdots \\ ka_{m1} & \dots & ka_{mn} \end{bmatrix}$$

- Many obvious properties:

$$A+B = B+A \qquad k(A+B) = kA+kB \qquad (A+B)^T = A^T+B^T$$

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Example

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 2 & 3 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 & 1 \\ 3 & 1 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$$

- $A + B =$
- $-2A + B =$
- $A + C =$

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- $A + B =$
- $-2A + B =$
- $A + C =$ cannot be performed!

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Product of matrices and linear equations

- The product of two matrices arises naturally from a system of linear equations:

$$\begin{cases} x_1 + 3x_2 + x_3 = 1 \\ 2x_1 - x_2 + x_3 = 1 \end{cases}$$

is a system of two linear equations with three unknowns x_1, x_2, x_3; in matrix form, the same system is written as two dot products:

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$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \Rightarrow \underbrace{\begin{bmatrix} \dots & R_1^T & \dots \\ \dots & R_2^T & \dots \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{b}.$$

$$\Rightarrow \underbrace{\begin{bmatrix} R_1^T x \\ R_2^T x \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{b}.$$

- Note $R_i^T x = R_i \cdot x$ the dot product

Row interpretation

- $Ax = y$ can be expressed as

$$R_i^T x = y_i \qquad \text{for every } i$$

where $R_1^T, R_2^T, \ldots R_m^T$ are rows of A.

- so $Ax = y$ is a 'batch' dot product of all rows of A with x

$$Ax = \begin{bmatrix} R_1^T x \\ R_2^T x \\ \vdots \\ R_m^T x \end{bmatrix} = y$$

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Column interpretation: linear combinations of the vectors

- Ax can be expressed as a linear combination of the columns of A with coefficients $x_1, \ldots, x_n$.

$$x_1 A_1 + x_2 A_2 + \ldots + x_n A_n$$

where $A_1, \ldots, A_n$ are the columns of A

- eg.

$$\underbrace{\begin{bmatrix} 1 & 3 & 1 \\ 2 & -1 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \begin{bmatrix} x_1 + 3x_2 + x_3 \\ 2x_1 - x_2 + x_3 \end{bmatrix}$$

$$= x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{A_1} + x_2 \underbrace{\begin{bmatrix} 3 \\ -1 \end{bmatrix}}_{A_2} + x_3 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{A_3}$$

- What is $Ae_1 = ?$

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Matrix-matrix multiplication

- the number of columns of the matrix to the left has to be equal to the number of rows of the matrix to the right.

$$A = \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}_{2\times 3} \text{ and } B = \begin{bmatrix} s_1 & t_1 \\ s_2 & t_2 \\ s_3 & t_3 \end{bmatrix}_{3\times 2}$$

$$AB = \begin{bmatrix} u^T s & u^T t \\ v^T s & v^T t \end{bmatrix}_{2\times 2}$$

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$$AB = \begin{bmatrix} u^T s & u^T t \\ v^T s & v^T t \end{bmatrix}_{2\times 2}$$

- Compute: $\begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 1 & 0 \\ 1 & -1 \end{bmatrix} =$

Some properties

- $A_{m\times n}B_{n\times \ell} = C_{m\times \ell}$;
- the product of matrices is not commutative: $AB \neq BA$.
- $AI = A$ also $IA = A$

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Invertible Matrices

A $n \times n$ matrix A is invertible or nonsingular if there exists a $n \times n$ matrix B such that

$$AB = BA = I.$$

- if A is nonsingular, its inverse is denoted as A^{-1};
- if A^{-1} exists, it undoes what A does.

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Inverse of a 2×2 Matrix

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $|A| = ad - bc \neq 0$, A^{-1} exists:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Example